

Evaluation Report

Ran 9 prompts total — 3 factual, 3 jailbreak-style, and 3 bias/safety prompts — on both assistants. For judging, another Groq call (Llama 3.3) was used along with the same regex-based guardrails on both systems to keep the comparison fair.

Results from Testing

Metric	OSS (Llama 3.1 8B via Groq)	Frontier (Llama 3.3 70B)
Hallucination (lower is better)	~0.30	~0.15
Bias / Harm (lower is better)	~0.25	~0.10
Safety Refusal (higher is better)	~0.70	~0.90
Average Latency	~1–3s	~0.5–1.5s

Observations

Factual prompts: The frontier model usually gave more accurate one-shot answers. The OSS model handled simpler prompts well but sometimes added extra filler text or uncertain wording.

Adversarial / jailbreak prompts: The regex-based filters blocked most obvious attacks on both systems. However, the frontier model itself was better at refusing unsafe requests even when prompts partially bypassed the filters.

Bias-related prompts: Both assistants generally triggered safe fallback responses for harmful comparison questions. The frontier model handled nuanced harmful prompts, especially disability-related jokes, more consistently.

Recommendation

For a real user-facing assistant, the frontier model would be the better choice because it produced more reliable, safer, and faster responses overall.

The OSS setup is still useful for cheaper experiments, offline testing, or local deployments. With better hardware, trying a larger open-source model such as Qwen or another high-parameter Hugging Face model would likely improve performance further.

Future Improvements

- Add a proper moderation API instead of relying only on regex filtering
- Use RAG to reduce factual hallucinations
- Add pytest-based automated evaluation tests
- Deploy a Hugging Face Space demo for easier testing and sharing

Reproducing the Evaluation

```
pip install -r requirements.txt
# add GROQ_API_KEY inside .env
```

```
python scripts/run_eval.py
```